



DEPARTMENT OF ELECTRICAL AND INFORMATION ENGINEERING

UNIVERSITY OF RUHUNA

Module Information Sheet

Code	EC4307	Title	Web Application Development			
Credits	3	Total Lectures (Hr)	30	Core/TE/GE	Core	
GPA/NGPA	GPA	Workshop (Hr)	24	Pre-requisites	None	
Aim		The aim is to equip students with the knowledge and skills required to design and develop modern web applications using industry-standard tools, frameworks, and best practices.				
Learning Outcomes		Upon successful completion of this module, students will be able to: LO-1 Describe the core components of web applications, including front-end, back-end, and database layers. LO-2 Demonstrate the ability to use web development frameworks and tools to build functional web applications. LO-3 Integrate various technologies (e.g., RESTful APIs, authentication, and security mechanisms) to create scalable and secure web applications. LO-4 Design a full-stack web application, considering performance, usability, and scalability, while justifying design decisions based on industry standards.				
Outline		Introduction to Web Development: Overview of web technologies, CSS Frameworks (Bootstrap, Tailwind CSS), JavaScript Frameworks (React.js, Angular.js), Client-server architecture, (HyperText Transfer Protocol) HTTP protocol. Full Stack Web Frameworks: Overview of Full Stack Frameworks (Frontend and Backend Development), Full Stack Frameworks (ASP.NET Core, Spring Boot, MERN (MongoDB, Express.js, React.js, and Node.js) Stack). RESTful Web Services: Introduction to RESTful APIs, REST Principles, Implementing RESTful Web Services Authentication and Authorization: Understanding Authentication vs. Authorization, Introduction to JSON Web Token (JWT)				
Methods of Assessments and Marks Allocation		● Continuous Assessments				

Eligibility to Appear for the End Semester Examination	• Minimum of 80% attendance for theory classes • Completion of all components of Continuous Assessments
Necessary Conditions to Pass the Module	• Earn at least 35% (i.e. 14 marks) of marks for Continuous Assessments • Achieve at least 35% (i.e. 21 marks) of the marks allocated for End Semester Examination
Recommended Readings	• Helland, V. M. Durano, J. Chilberto, and E. Price, <i>"ASP.NET Core 5 for Beginners"</i>, Packt Publishing, 2020. ISBN: 978-1-80056-718-4. • G. Lim, <i>"Beginning MERN Stack"</i>, 2021. ISBN: 979-8-52362-550-3. • R. Wieruch, <i>"The Road to Learn React: Your Journey to Master Plain Yet Pragmatic React.js"</i>, CreateSpace Independent Publishing Platform, 2017, ISBN: 9781979807074. • C. Walls, <i>"Spring Boot in Action"</i>, 1st ed., Manning Publications, 2016, ISBN-10: 1617292540.